NATURE INSPIRED STORAGE AND COMPUTING

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TABLE OF CONTENT

PROBLEM STATEMENT

- Conventional storage systems store data in binary as zeros and ones.
- Due to this, they face limitations as mentioned below:

    1. They have limited physical space.
    2. Do not last as long as the magnetic/semi-conductor components wear out.
    3. Consume enormous amount of energy and space when storing large amounts of data.

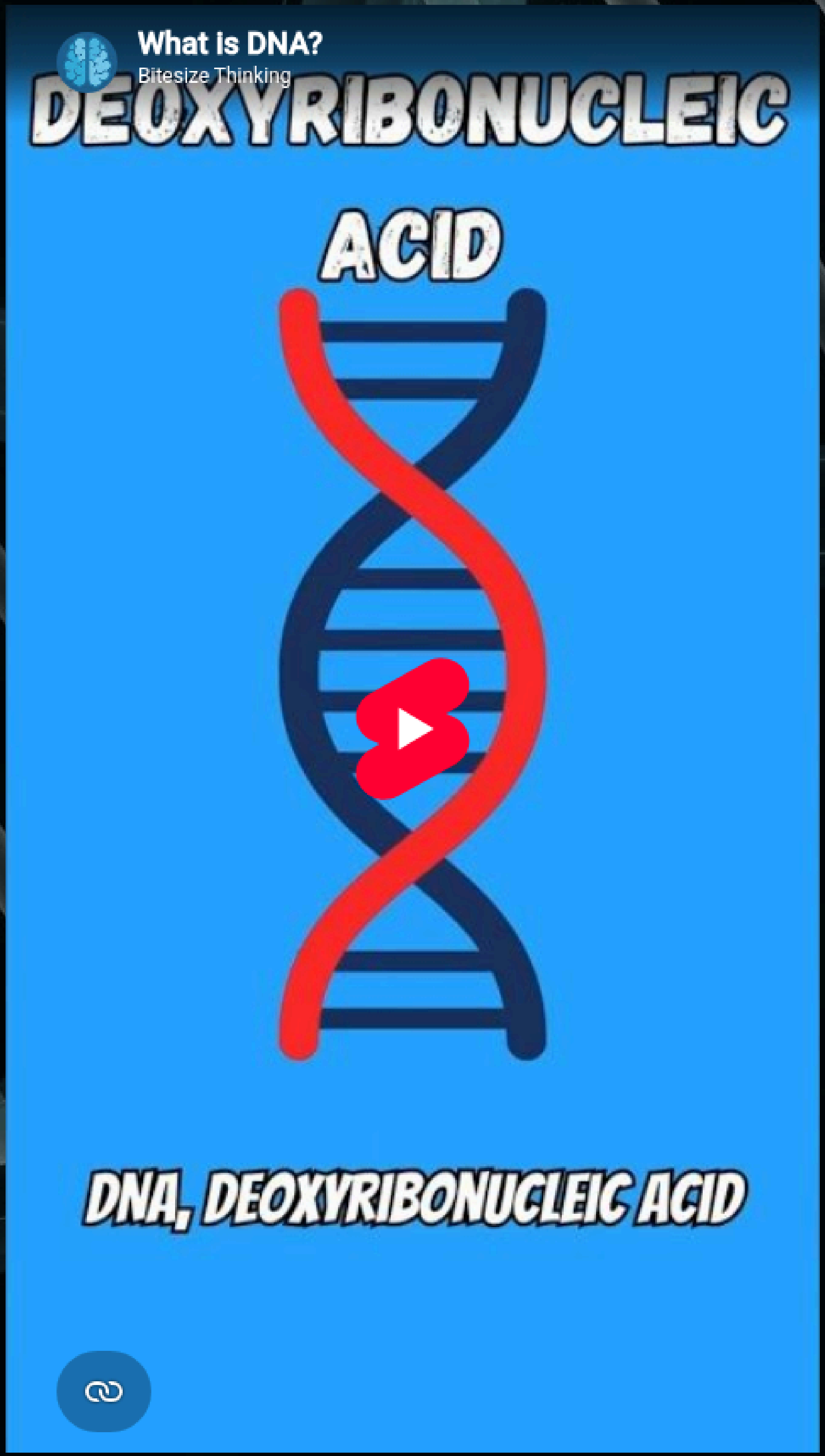
PROBLEM STATEMENT

- Similarly, with the rising requirements to store data, so does the need for equally advanced and efficient algorithms.
- In problems where a multitude number of solutions may be possible, conventional systems execute every solution one by one and proceed making the process highly inefficient in bigger datasets.

- With the exponential growth of digital data and the limitations of conventional silicon-based storage and computing technologies, bio-inspired, DNA-driven computing and information storage has emerged as a promising solution.

WHAT IS DNA?

- DNA is structured as a double helix, with each side composed of repeating units called nucleotides.
- Each nucleotide contains one of four chemical bases: Adenine (A), Thymine (T), Cytosine (C), and Guanine (G).
- The sequence of these bases forms a chemical code that carries genetic instructions.
- The DNA molecule's double-helical structure and chemically stableness is why the info remains preserved for long periods under suitable conditions in a very compact manner.



WHY DO WE STORE DNA?

We store digital data in DNA because DNA is extremely dense, durable, and suitable for long-term archival storage. DNA naturally stores biological information using four bases and scientists can similarly encode digital information such as text, images, audio, and videos into DNA sequences.

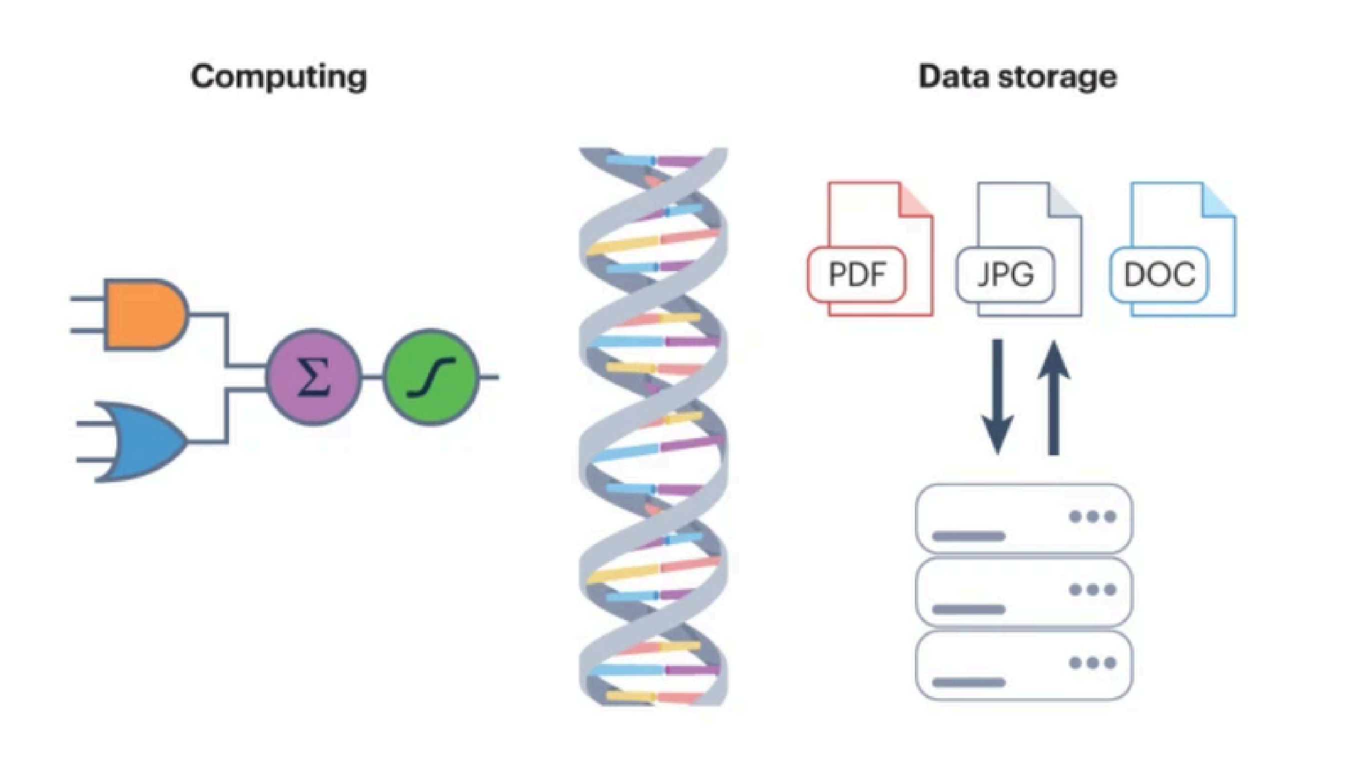
The Main Reasons are :

- Very high storage capacity
- Long-term stability
- Low maintenance for archives
- Capacity to store different types of files

WHAT IS DNA STORAGE?

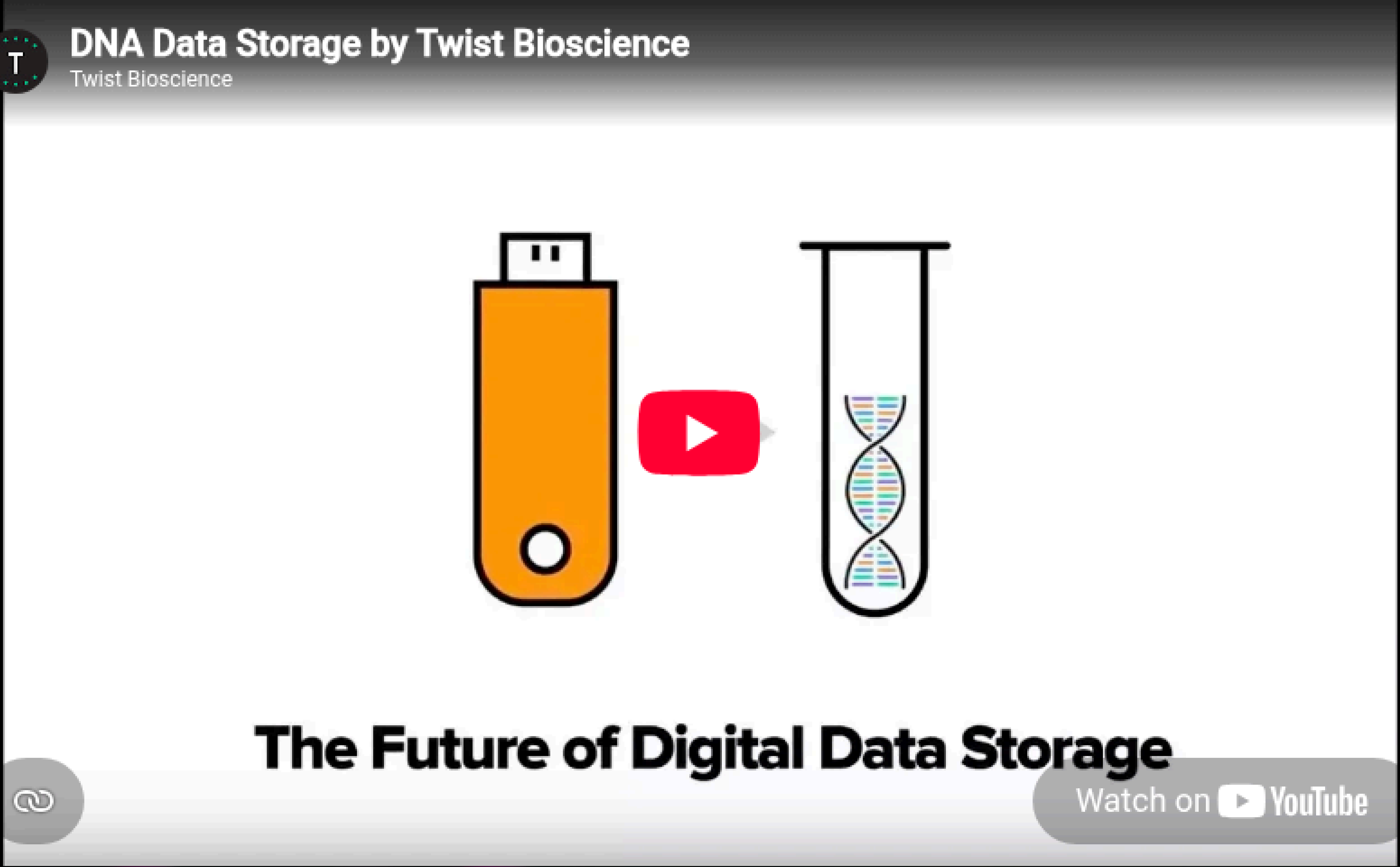
DNA storage, or DNA-based data storage, is a method of storing digital information—such as text, images, audio, and videos—in artificially synthesized DNA molecules instead of conventional devices like hard drives or flash drives.

Simple working process:
Digital file → Binary code (0 and 1) → DNA sequence → DNA synthesis and storage → DNA sequencing → Decoding → Original file



HOW DOES DNA WORK?

- Cells can retrieve and use genetic information through processes such as transcription and translation.
- During transcription, a particular DNA sequence is copied into messenger RNA, which can then be used to produce proteins and continue their processes.
- This phenomenon is mimicked to make dna inspired storage systems.
- Flow: information is encoded into a sequence → the sequence is physically stored → the sequence is copied or preserved.
- The information is later read back, the synthetic DNA is sequenced, and computational algorithms decode the nucleotide sequence back into the original digital file


DNA Data Storage by Twist Bioscience
Twist Bioscience
The Future of Digital Data Storage
Watch on YouTube

BIOLOGICAL PHENOMENON GENETIC ALGORITHMS

- Genetic algorithms are inspired by evolution through natural selection.
- Mutations and genetic recombination create differences.
- Over many generations, these traits become more common in the population.
- Instead of organisms, the algorithm works with a population of possible solutions.
- Each solution is represented as a chromosome, and a fitness function evaluates how well it solves the problem.
- Their characteristics are combined through crossover, and small random changes are introduced through mutation, thus defiining the most efficient solutions.

APPLICATIONS

1. Long-Term Digital Archiving

- DNA can be used to preserve large amounts of digital information for very long periods.
- It is particularly suitable for cold/archive data that does not need to be accessed frequently.
- Potential users include libraries, museums, research institutions and large data archives.
- DNA's high information density and stability make it attractive as a future archival medium.

Visual: Digital data → DNA strands → Long-term archive

APPLICATIONS

2. Storing Massive Digital Datasets

- DNA can store text, images, audio, video and other digital files.
- Researchers have demonstrated systems capable of storing hundreds of megabytes of data in DNA.
- DNA storage can also support random access, allowing a particular file to be retrieved without reading the entire DNA archive.
- This could be useful for future large-scale data centres and archival databases.

Visual: Multiple files → DNA database → Selected file retrieved

APPLICATIONS

3. Searchable & Dynamic DNA Data Storage

- DNA storage is moving beyond simply writing and reading files.
- Researchers have demonstrated ways to access specific DNA sequences and perform file operations.
- Molecular addressing can act like an index for finding particular information.
- This opens possibilities for searchable molecular databases and more flexible storage systems.

Visual: Object/material → embedded DNA → information recovered

APPLICATIONS

4. Optimizing DNA Storage

Genetic Algorithms (GAs) can be used as optimization techniques alongside DNA storage systems. Possible applications include:

- Optimizing DNA encoding sequences.
- Finding sequences that satisfy biological constraints such as suitable GC content and reduced unwanted sequence patterns.
- Improving the reliability and efficiency of DNA data storage.
- Searching through a large number of possible solutions to identify better encoding strategies.

Visual: Genetic Algorithm → candidate DNA sequences → selection/optimization → better DNA code

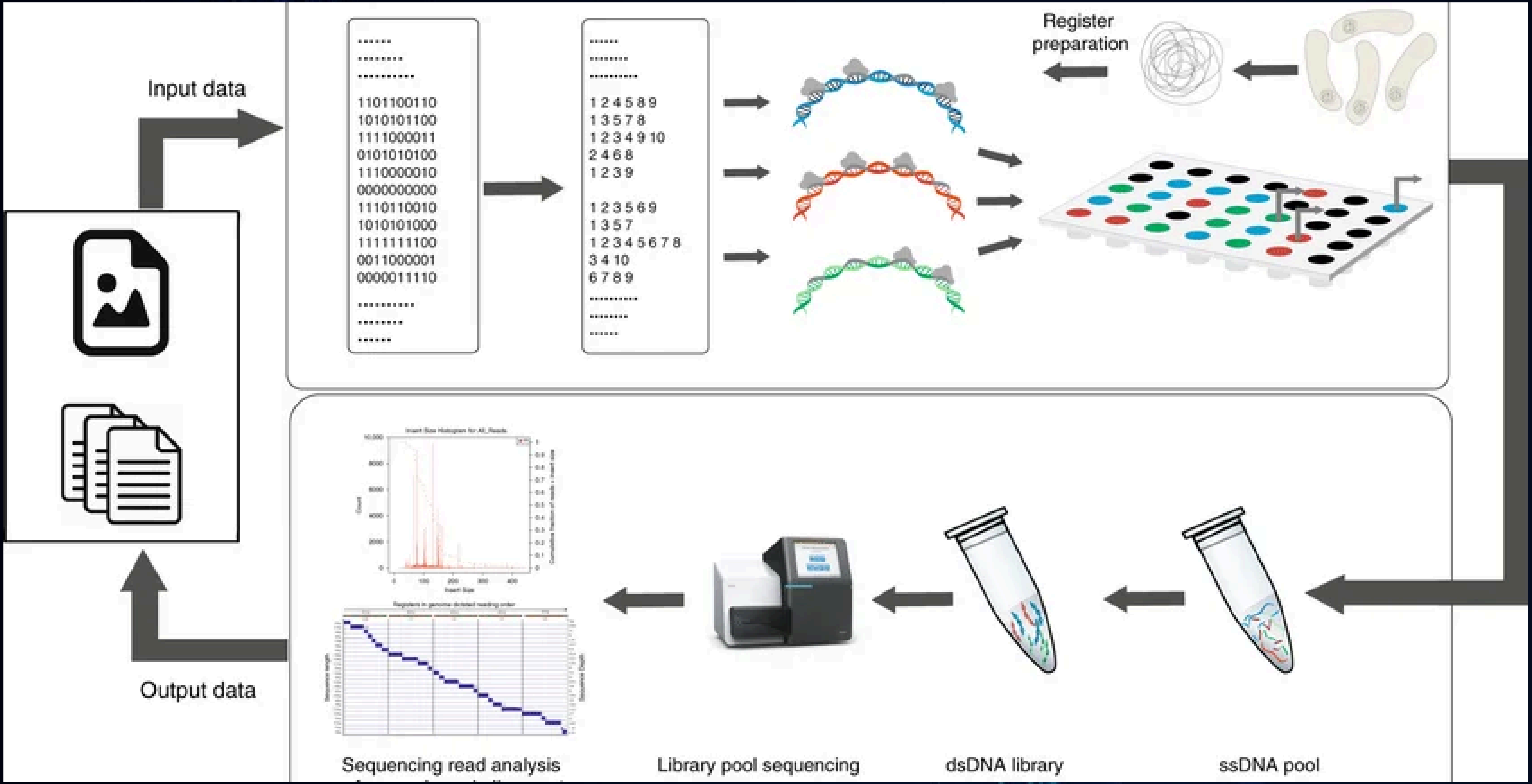
APPLICATIONS

5. Bioinformatics & Biological Data Analysis

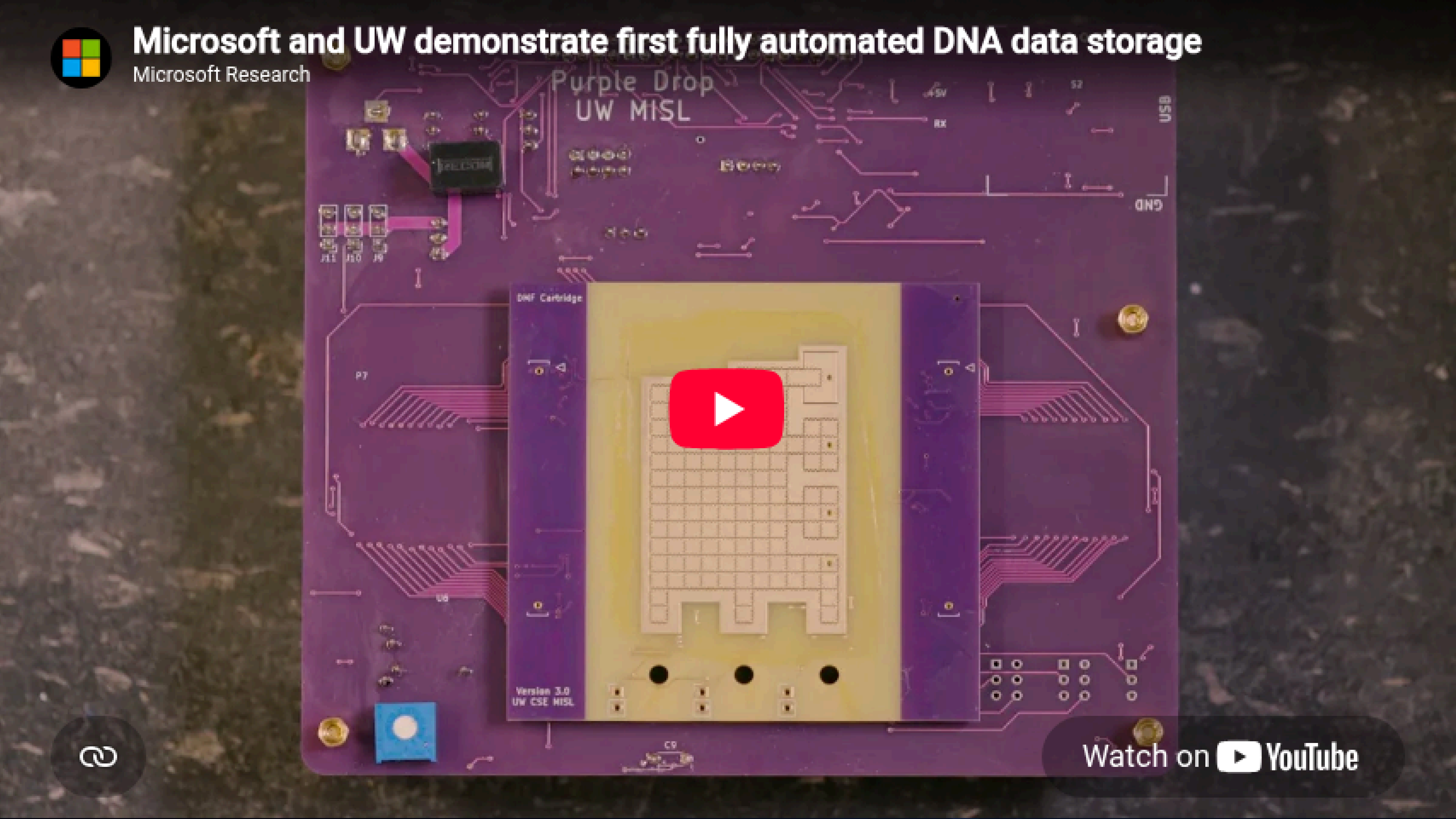
- Genetic algorithms solve complex bioinformatics optimization problems.

- Applications include DNA/protein sequence analysis, multiple sequence alignment, and protein structure prediction.

- They can help search large solution spaces where finding the best solution directly is difficult.



-APPLICATIONS ENHANCED BY DNA-BASED DATA STORAGE-


Microsoft and UW demonstrate first fully automated DNA data storage
Microsoft Research
Purple Drop
UW MISL
DMF Cartridge
Version 3.0
UW CSE MISL
Watch on YouTube

PROS AND CONS

PROS

DNA-Based Data Storage

- High density: Stores huge data in tiny space.
- Long-lasting: Preserves data for long periods.
- Low maintenance: Needs less energy and infrastructure.
- Random access: Retrieves specific files easily.

Genetic Algorithm

- Complex optimization: Solves difficult problems efficiently.
- Flexible: Works for many optimization problems.

CONS

DNA-Based Data Storage

- High cost: Expensive to create and read.
- Slow: Writing and retrieval take time.
- Errors: Risk of data loss or mistakes.
- Complex: Needs specialized equipment.

Genetic Algorithm

- Premature convergence: May find a poor solution early.
- High computation: Needs significant processing power.

RESEARCH TIMELINE

1975

Genetic Algorithms- John Holland

Established the Genetic Algorithm (GA) as a computing approach inspired by natural selection and evolution.

1994

DNA Computing- Leonard Adleman

Demonstrated that DNA molecules could be used to perform computation, famously solving a combinatorial problem. This became a foundation of DNA-based computing.

2001

Long-term information storage in DNA- Bancroft et al.

Demonstrated the potential of DNA for long-term information preservation, helping establish DNA as a storage medium.

RESEARCH TIMELINE

2012

DNA Data Storage- Church, Gao & Kosuri

Demonstrated large-scale digital information storage in DNA, encoding a 5.27-megabit book and recovering it through DNA sequencing.

2013

Practical DNA Storage- Goldman et al.

Improved DNA storage using coding and redundancy, moving the field toward high-capacity and reliable archival storage

2017

DNA Fountain- Erlich & Zielinski

Introduced a highly efficient coding architecture and stored approximately 2.14 MB of digital information in DNA.

RESEARCH TIMELINE

2018

Random-Access DNA Storage — Organick et al.

Stored 35 files (>200 MB) and demonstrated retrieval of individual files without sequencing the entire DNA pool.

2023

Evolutionary Optimization for DNA Storage — Rasool et al.

Applied an evolutionary approach to construct robust DNA storage codes, directly linking nature-inspired computing with DNA data storage.

STATISTICAL DATA

Parameter	DNA-Based Data Storage	Conventional Data Storage
Experimental storage density	215 PB/g	Conventional media are typically far lower on a mass basis
Advanced experimental density	17 EB/g	Much higher than conventional physical storage media
Theoretical storage density	Up to approximately 455 EB/g	Far below DNA's theoretical molecular density
Data retention / longevity	Thousands of years under suitable preservation	Digital systems often require replacement/migration over decades
Error-free recovery	Demonstrated experimentally	Depends on device condition and error-correction systems
Storage maintenance	Potentially very low during passive archival storage	Requires active infrastructure and periodic maintenance/migration
Access/read speed	Currently much slower	Very fast, typically nanoseconds to milliseconds depending on medium
Best application	Cold/archival data stored for years or centuries	Frequently accessed, everyday data

STATISTICAL DATA

Parameter	Statistical Data
Maximum information per DNA base	2 bits
Demonstrated data storage	200+ MB
DNA strands used	13+ million
Random-access files stored	35 files
Physical storage density	295 PB/g
Theoretical storage density	Up to 17 EB/g
Estimated storage lifetime	~20,000 years at 9.4°C

DNA-BASED STORAGE SYSTEMS

Storing data in the code of life for a more sustainable future.

- A Adenine
- T Thymine
- C Cytosine
- G Guanine

WHY DNA STORAGE?

Ultra-high density
1 gram of DNA can store ~215 petabytes of data.

Long durability
DNA can last thousands of years.

Low energy
Requires minimal energy to store and preserve.

Environmentally friendly
Biodegradable and non-toxic storage medium.

HOW IT WORKS

1. DIGITAL DATA → **2. BINARY CONVERSION** → **3. ENCODING** → **4. DNA SYNTHESIS** → **5. STORAGE**

1. DIGITAL DATA: Any type of data (text, images, videos, databases, etc.)
2. BINARY CONVERSION:

   01010110 01010010
   10110011 01010100
   01010100 10101010

   Data is converted into binary code (0s & 1s).
3. ENCODING:

   A T G C A
   C G A T C
   T A C G T

   Binary is mapped to DNA bases (A, T, C, G).
4. DNA SYNTHESIS: The DNA sequence is synthesized in a lab.
5. STORAGE: DNA is stored in tiny vials in stable conditions (dry, dark, cool).

RETRIEVAL & DECODING

4. ORIGINAL DATA ← **3. DECODING** ← **2. SEQUENCING** ← **1. RETRIEVE DNA**

4. ORIGINAL DATA: Reconstruct the original digital data.
3. DECODING:

   A T G C A
   C G A T C
   T A C G T

   Convert the DNA sequence back to binary code.
2. SEQUENCING: Determine the order of nucleaotides (A, T, C, G).
1. RETRIEVE DNA: Retrieve the DNA sample from storage.

STORAGE DENSITY COMPARISON

Hard Drive	USB Drive	DNA (1 gram)
~1 TB	~64 GB	~215 PB

1 gram of DNA can store
~215 MILLION GB
of data!

APPLICATIONS

- Long-term archival storage
- Space missions & deep space data
- Medical records & genomic databases
- Preserving cultural heritage
- Secure & tamper-proof data preservation

"DNA is nature's hard drive. We're just learning how to write to it."

SUMMARY

FUTURE POSSIBILITIES

DNA-Based Data Storage

- Massive archival storage
- Long-term digital archives
- Lower-energy storage
- Faster & cheaper systems
- Integration with biotechnology

FUTURE POSSIBILITIES

Genetic Algorithms

- Smarter optimization
- Bioinformatics & sequence design
- Hybrid AI systems
- Engineering & design